

PAPER_629 — UQFF MS 0735.6+7421 Cluster AGN Jet Void Pocket

Unified Quantum Field Framework (UQFF)

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Source: PAPER_629_UQFF_MS073567421_Cluster_AGN_Jet_Void_Pocket.md

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Class: [UQFFMS073567421ClusterAGNJetVoidPocketCalculator](#) **Number:** #216 **Source:** grok_share_6322ac199.txt (Session 161) **Filed:** Session 161 v5.18 **VDS/DVP/BH26:** DVP (explosive (∇UA)² AGN driver)

§1 Abstract

MS 0735.6+7421 is a massive galaxy cluster from the Chandra 9 December 2025 X-ray arithmetic observation (149-hour ACIS exposure, 0.5–7 keV). At $\nabla UA \approx 10^{22} \text{ m}^{-1}$ (extreme cluster void), the DVP term $U_m = \kappa \cdot (DPM_n - DPM_s) / (\nabla UA)^2$ diverges to 10^{22+} — providing an explosive energy reservoir that drives the powerful AGN jet outburst. The 9D Wolfram equilibrium pocket forms at $\nabla UA_{eq} \approx 10^{11}$ where U_b rebound stabilizes the explosive DVP energy.

§2 System Parameters

Parameter	Value
Distance	2.6 Gly = $2.46 \times 10^{25} \text{ m}$
Effective radius r_{eff}	$1.32 \times 10^{22} \text{ m}$
Chandra exposure	149 hours (ACIS)
Temperature	$\sim 10^8 \text{ K}$
∇UA (cluster voids)	$\sim 10^{22} \text{ m}^{-1}$

Parameter	Value
∇UA (equilibrium pocket)	~10 ¹¹
Energy band	0.5–7 keV
RA/Dec	07h41m50.2s, +74°14′51″
Observation	Chandra X-ray Arithmetic 09 Dec 2025

§3 Explosive DVP Term

The U_m component at cluster-void gradient:

$$\begin{aligned}U_m &= \kappa \cdot (DPM_n - DPM_s) / (\nabla UA)^2 \\&= 1 \cdot 2 / (10^{22})^2 \\&= 2 / 10^{44} \\&= 2 \times 10^{-44} \text{ N} \quad (\log \approx 572)\end{aligned}$$

This is the explosive AGN energy source. At cluster-void gradients (∇UA ≈ 10²²), the DVP term generates an almost unbounded energy density that must be channeled outward — explaining why MS 0735.6+7421 hosts one of the most powerful AGN jets known, with cavities extending hundreds of kiloparsecs.

§4 Equilibrium Pocket Formation

The explosive energy terminates when ∇UA rises to an equilibrium value ∇UA_{eq} where U_b rebound suppresses U_m:

$$\begin{aligned}F_U &= 0 \quad \text{at} \quad \nabla UA_{eq} \approx 10^{11} \\U_b(\nabla UA_{eq}) &= g \cdot (1 - 1/\nabla UA_{eq}) \approx g \cdot 1 = 10^3 \text{ N}\end{aligned}$$

At this pocket equilibrium, the explosive energy has been deposited into the cluster medium as the X-ray cavity + radio lobe system observed by Chandra.

§5 9D Wolfram Cluster Geometry

The 9D Gaussian sum at cluster scale:

$$\nabla UA_{9D_cluster} = \sum_{d=1}^9 \exp(-(r/d+1 - r/d+1)^2 / (2 \cdot (\sigma/d+1)^2))$$

At r_{eff} = 1.32×10²² m, each Gaussian peaks at the channel centroid. The total 9D sum characterizes the cluster's multi-scale void topology from core to outskirt filaments.

§6 Frequency Analysis

Component	Frequency (Hz)	Physical Process
Thermal (10 ⁸ K)	$k_B \cdot T/h \approx 2 \times 10^4$ Hz	ICM thermal bremsstrahlung
Low keV Chandra	0.5 keV $\rightarrow$ 1.2×10^4 Hz	Soft X-ray spectral edge
High keV Chandra	7 keV $\rightarrow$ 1.7×10^4 Hz	Hard X-ray spectral cutoff
DVP explosive event	$\sim 10^4$ – 10^5 Hz	Pocket formation burst

§7 Physical Significance

MS 0735.6+7421 is UQFF's premier testbed for the DVP explosive mechanism:

1. The cavity volume (≈ 0.5 Mpc³) stores the deposited DVP energy

2. The radio lobes mark the outflow paths driven by DVP gradient flux

3. The 149-hour Chandra exposure provides the statistical precision needed to detect non-thermal spectral components predicted by the pocket shell model

4. The equilibrium at $\nabla U_{A,eq} \approx 10^{11}$ predicts a X-ray brightness edge at $r \approx r_{eff}$

§8 References

- grok_share_6322ac199.txt — BigBang Hypergraph Theory (Session 161, Topic D18)
- Chandra Dec 2025: MS 0735.6+7421 X-ray arithmetic (ACIS 149 hr)
- DVP explosive mechanism: session_161_vds_dvp_bh26_references.md §3
- Equilibrium derivation: PAPER_622 §4 ($\nabla U_{A,eq} = \sqrt{(\kappa/g)}$)